Preliminary Results

Ryan Waterman

Table of Contents

Data Preparation and Feature Engineering

The data preparation and feature engineering process conducted during the initial experimentation can be broken into three subsections: data cleaning and preparation, feature engineering, and train-test data split. Each of these subsections are explained, in detail, below.

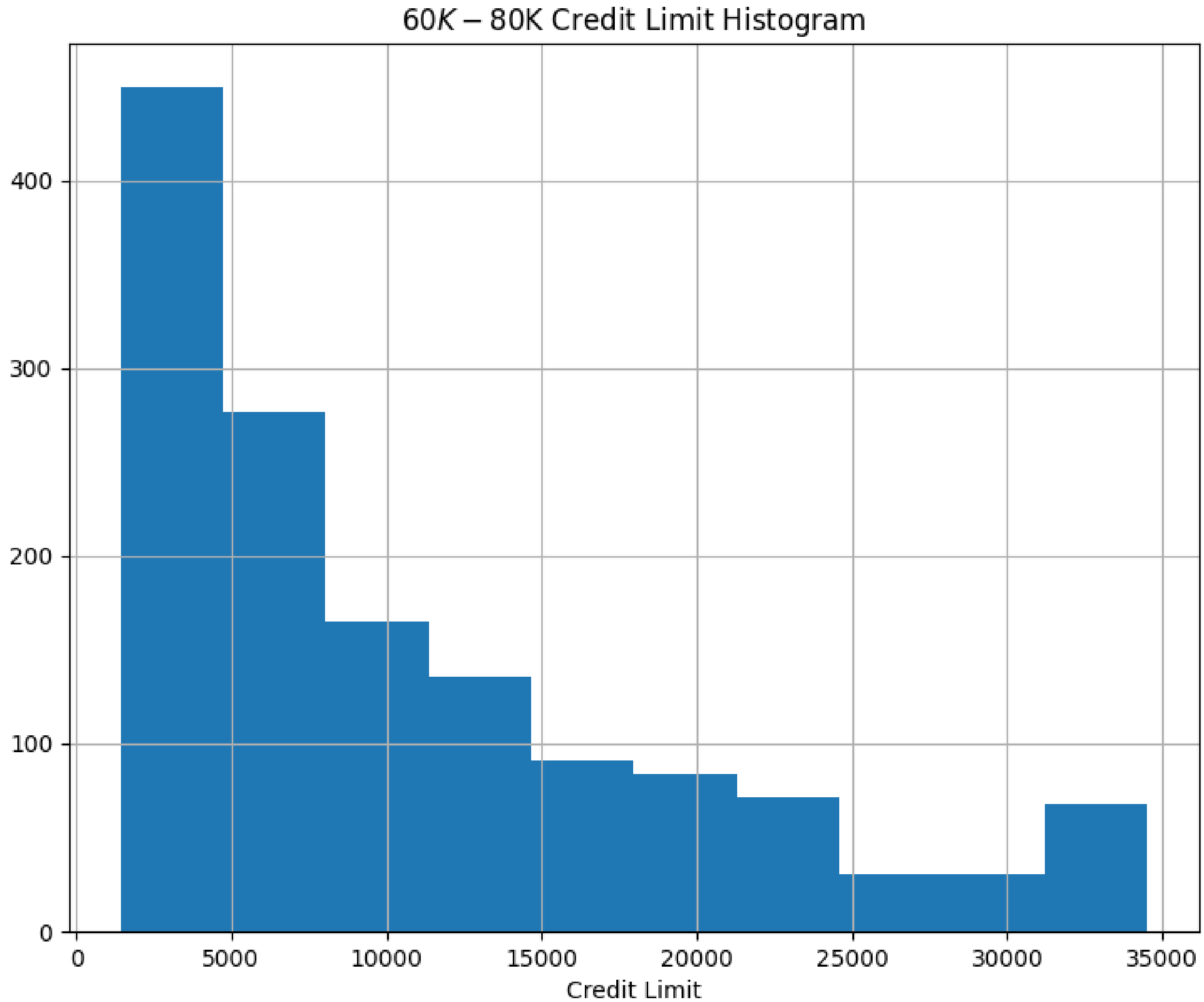
Data cleaning and preparation was performed experimentally throughout the exploratory model building process. Initially, all models (KNN, QDA, and Logistic Regression) were naïvely trained on the entire data set to understand the model performance with the unmodified data set. During this training process, class imbalance was identified as a key issue for model performance. This was corrected experimentally for each model with a cleaning process that randomly removed (random seed = 42) a given percentage of the observations where Attrition_Flag equaled 0. Each model was trained with configurations of no reduction in Attrition_Flag = 0 observations, 65% reduction of Attrition_Flag = 0 observations, and 75% reduction of Attrition_Flag = 0 observations. Additionally, stepwise variable selection was utilized to remove non-influential variables from the data set, as seen in the Robust Variable Selection section. This, in tandem with the class imbalance correction, encompasses the methodology for data preparation and cleaning used during this phase of model development.

Feature engineering represented a significant effort in this analysis. Harkening back to the Analytic Plan phase of this project, there were four new variables identified for feature engineering: Balance_To_TC, Credit_Limit_per_Income, Leveraged_Balance_Ratio, and Relationship_Strength. A brief description of each variable can be found, below:

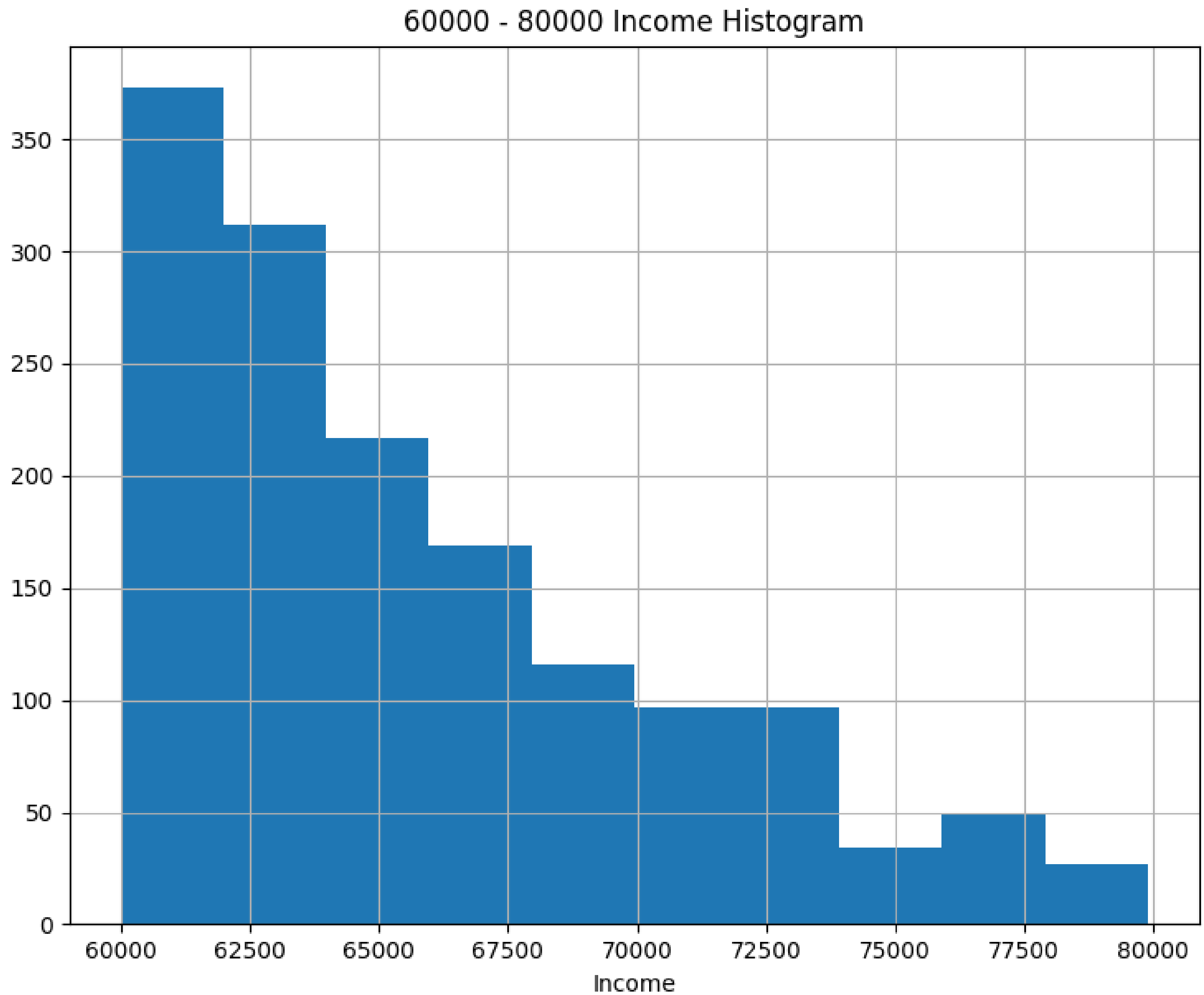
1. Balance_To_TC = (Total Revolving Balance) / (Total Transaction Count)
    a. A high revolving balance to transaction count ratio would imply the customer is spending large amounts with few transactions, or vice versa, which may have an impact on churn rate.
2. Credit_Limit_Per_Income = Credit Limit / Numerical_Income
    a. The income categories would be reduced to numerical representations. If customers have obtained a high credit limit relative to their income, it can be assumed they have high credit scores, for which the number of open accounts in good standing is a factor.
3. Leveraged_Balance_Ratio = (Total Transaction Amount) / (Average Utilization Ratio * Credit Limit)
    a. This feature is an indication of how capable the customer is of paying their balance, normalized to their credit limit.
4. Relationship_Strength = Months on Book * Total Relationship Count
    a. A representation of the strength of the relationship the customer has with the bank. A higher value indicates a stronger relationship, as customers that are heavily invested in the products and services the bank provides may be less likely to stop utilizing the bank.

A feature worth highlighting is Credit_Limit_Per_Income, which proved to be far more difficult to implement than initially thought. The Income_Category variable is categorical in nature, therefore requiring a conversion to a numerical format in order to implement this feature. The categories representing Income_Category are $60K - $80K, Less than $40K, $80K - $120K, $40K - $60K, $120K +, and Unknown.

In order to create a distribution of numerical incomes that fit the correct proportion of categorical income values, a key assumption was made; credit limit is proportional to income. Using this assumption, credit limit histograms grouped by income category were generated to understand the distribution of credit limits in each income category. An example of the $60k-$80k Income Category grouped Credit Limit histogram can be seen, below:



For each income category, the corresponding credit limit values were binned into 10 groups, for which the numerical Income Category distribution would be based, as the assumption is that credit limit and Income are proportional. For each income category, the credit limit bin was mapped to the income range of the category, then random values (random seed = 42) were generated within the corresponding income bin for the number of values represented in the credit limit distribution. This results in a numerical Income value that is mapped to the distribution of credit limits for the initial income range. To verify the results of this transformation, histogram plots were created for the respective, numerical income ranges, seen below for numerical incomes in the $60,000 to $80,000 range:



Upon the completion of generating the Numerical Income variable, the intended feature, Credit_Limit_Per_Income, could be generated by dividing the credit limit by the Numerical Income.

Lastly, the train-test split selected for the entirety of the analysis was 80% train, 20% test with a random seed of 42. Stratification was done prior to splitting data on the entirety of the data set, as detailed in the class imbalance correction methodology from the data cleaning and preparation section.

Model Implementation and Results

Model implementation was completed with 3 phases of variable selection: Naïve (all predictors in data set), Initial Selection (variables selected as part of the analysis in the analytical plan), and Stepwise Selection (stepwise variable selection with a combination of forward and backward methods). Each configuration of variables was used to evaluate Logistic Regression, QDA, and KNN models with various levels of class imbalance correction.

Null models were trained for each of the respective model types listed above. The Confusion Matrices, ROC Curves, and AUC scores can be seen in Images 1 through 3, below.

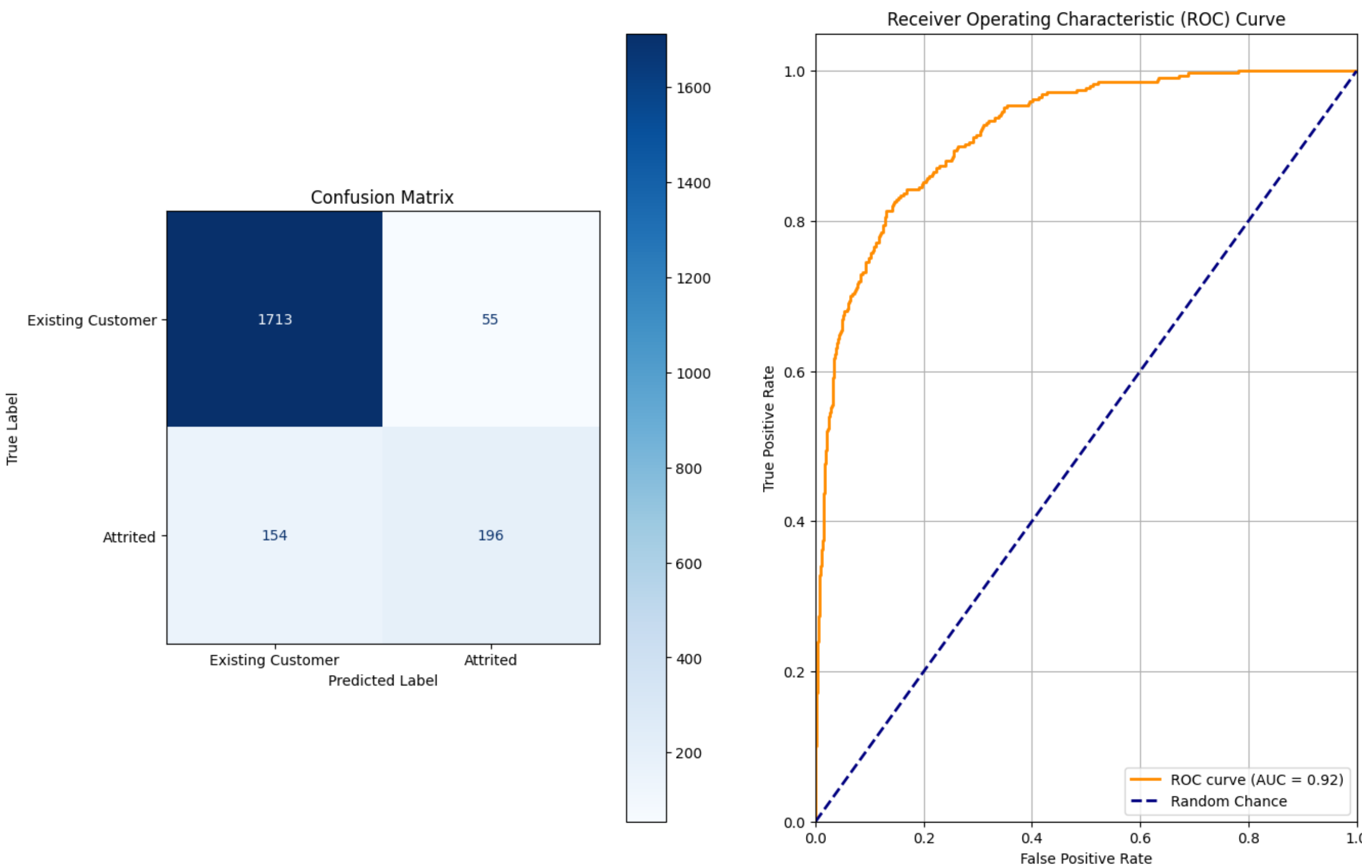


Image 1: Logistic Regression Null Model Confusion Matrix and ROC Curve

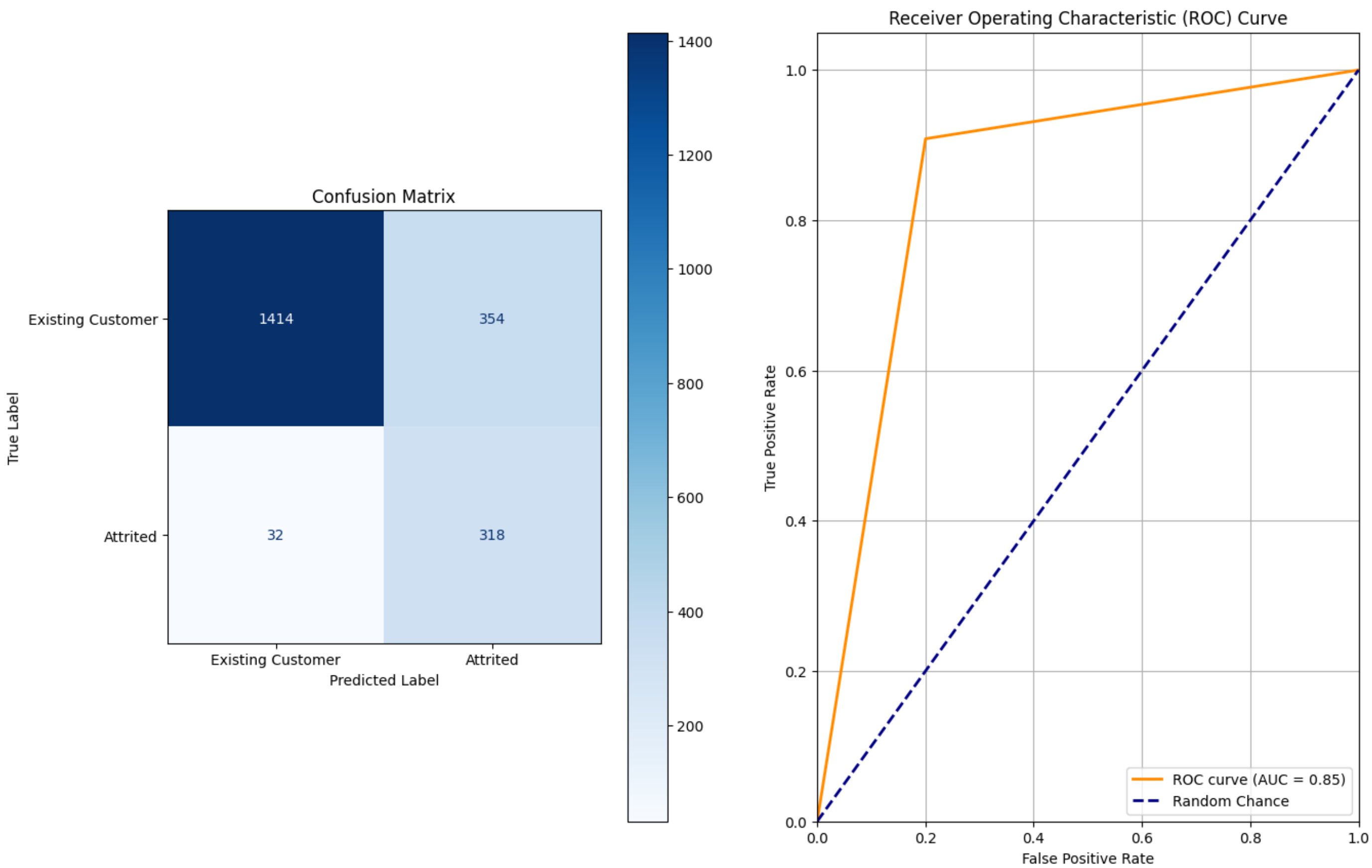


Image 2: QDA Null Model Confusion Matrix and ROC Curve

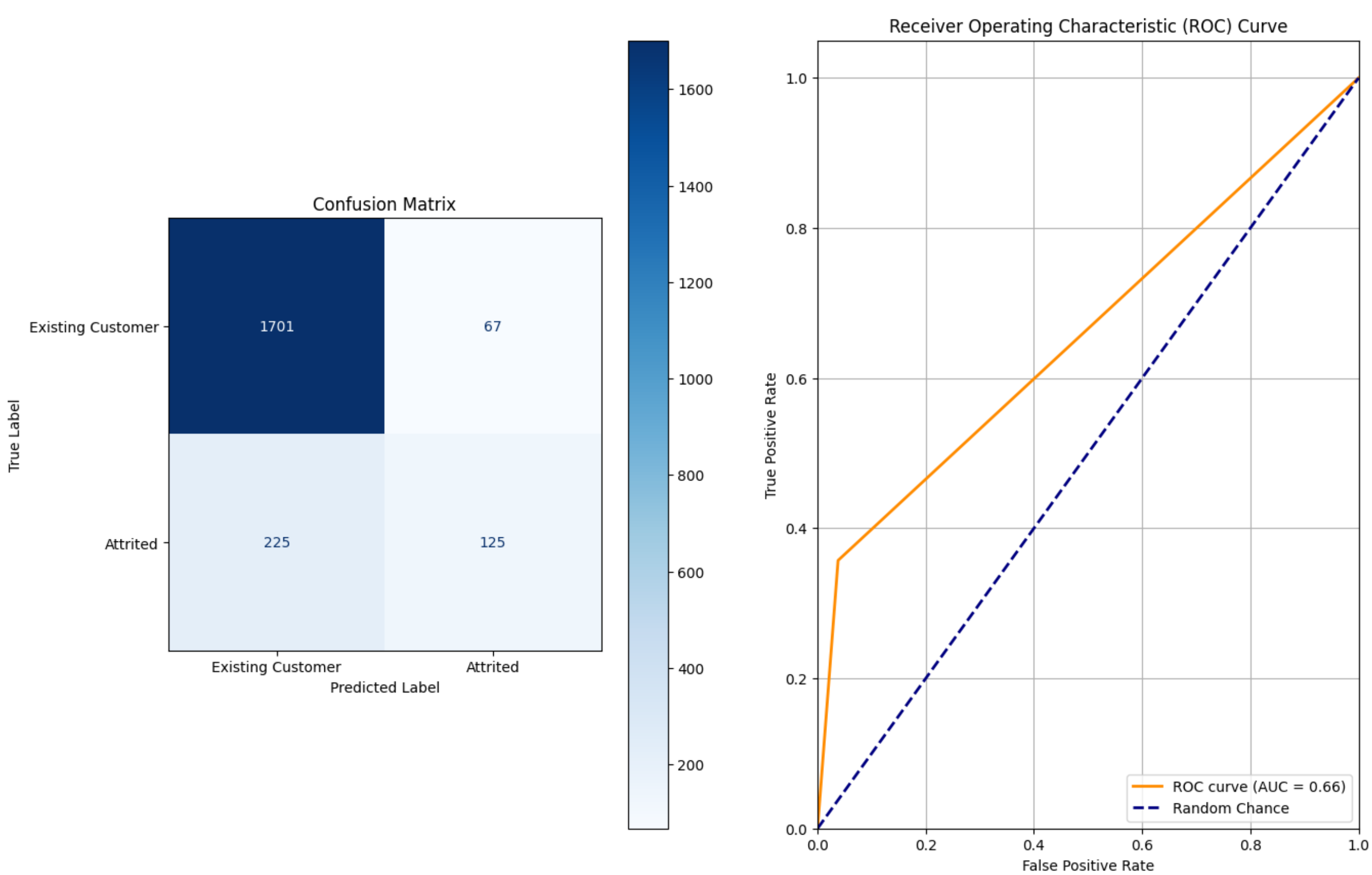


Image 3: KNN Null Model Confusion Matrix and ROC Curve

The ROC curve for the Logistic Regression model is smooth because of the continuous probability output from the model prediction. QDA and KNN models output more discrete probabilities, resulting in a sharp, jagged ROC curve.

Critical performance metrics for the Null Models are found in Table 1, below:

Model Type	Accuracy	Recall	Precision	F1-Score
Logistic Regression	90.13%	56%	78.09%	0.65
QDA	81.78%	90.86%	47.32%	0.62
KNN	86.21%	35.71%	65.1%	0.46

Table 1: Null Model Performance Metrics

By most metrics, Logistic Regression is the best performing null model, with the highest accuracy, precision, and F1-Score.

After optimizing feature selection and correcting for class imbalance, each model configuration's performance improved. The ROC Curves and AUC Scores for the optimized models can be seen in Images 4 through 6, below.

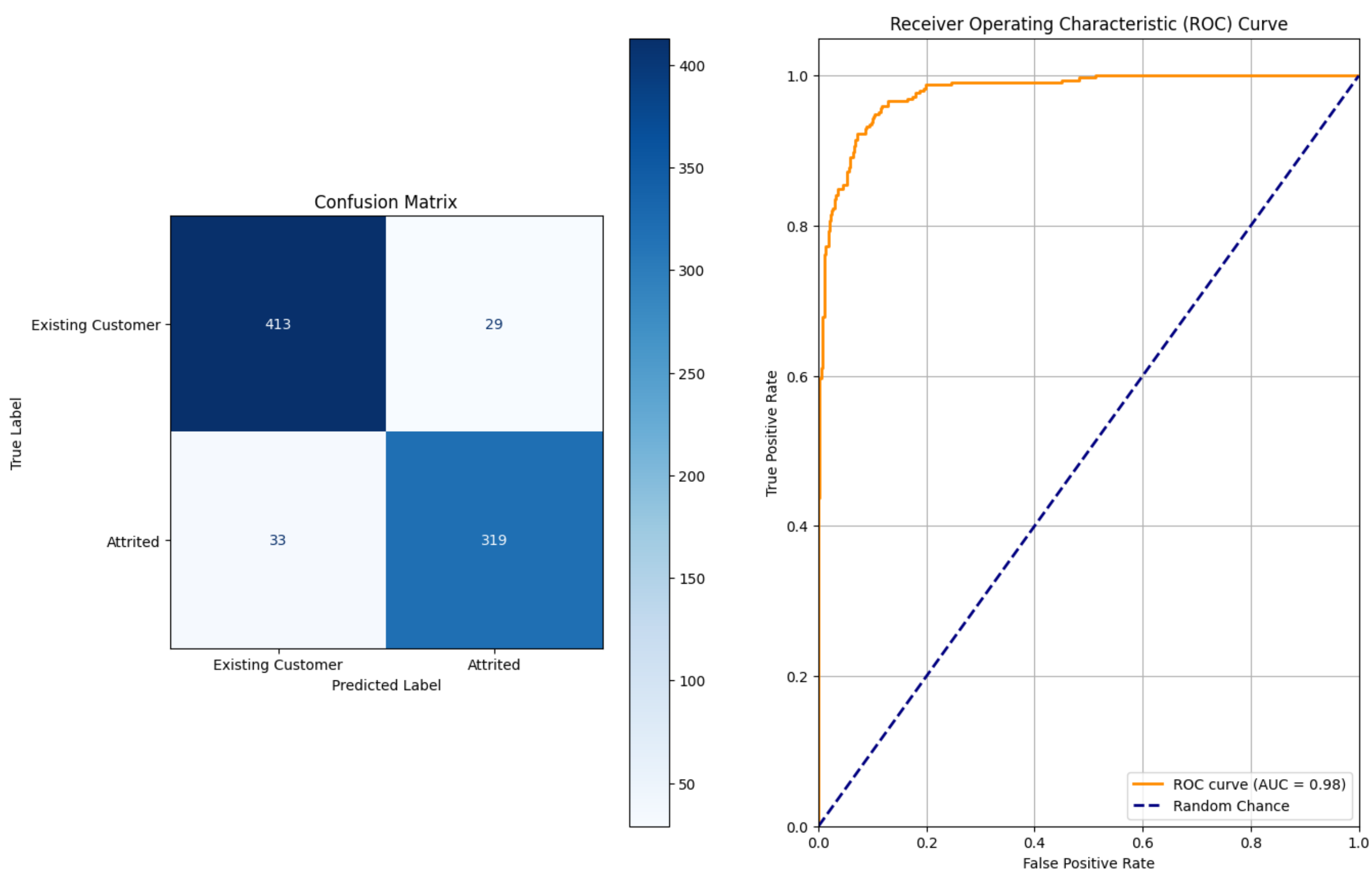


Image 4: Optimized Logistic Regression Model Confusion Matrix and ROC Curve

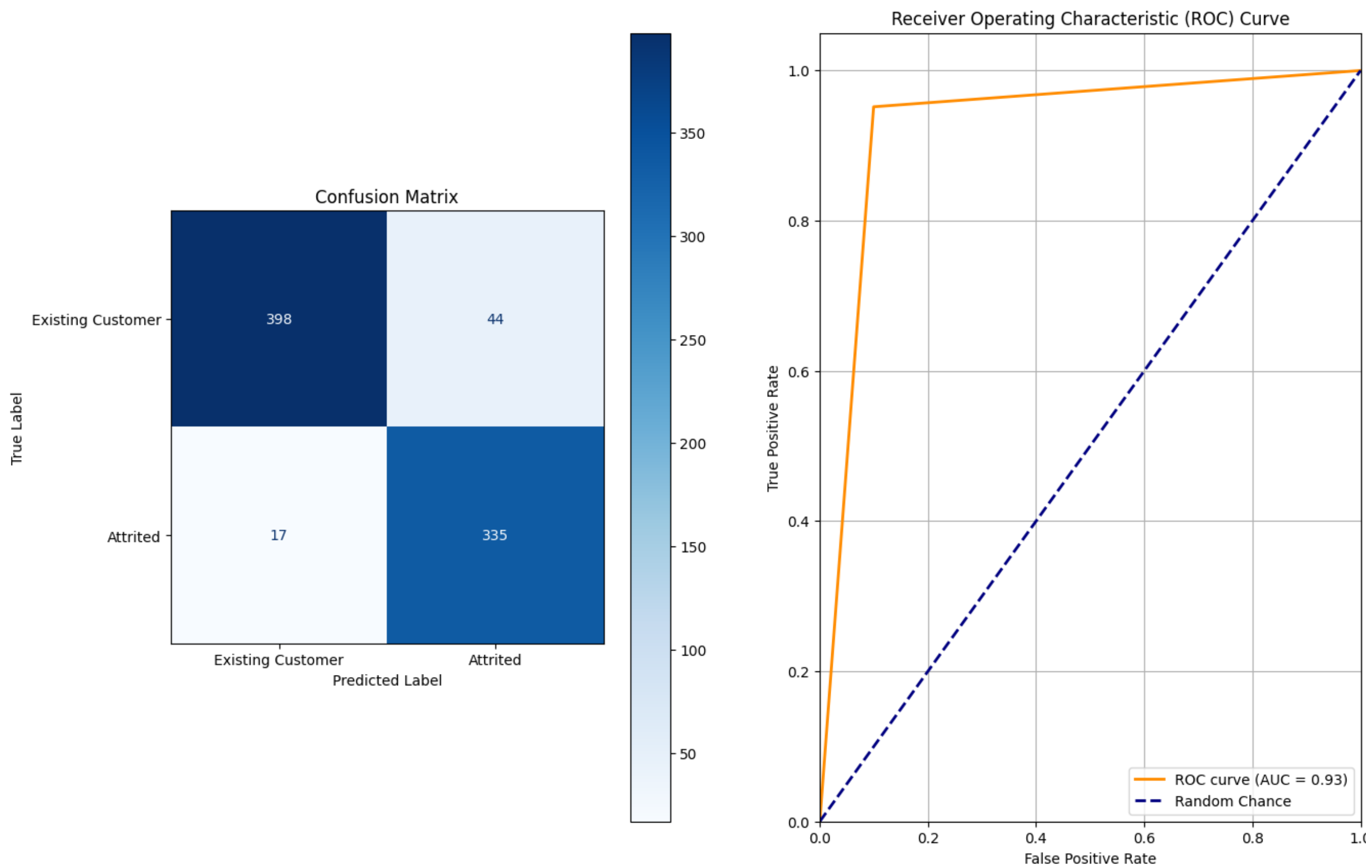


Image 5: Optimized QDA Model Confusion Matrix and ROC Curve

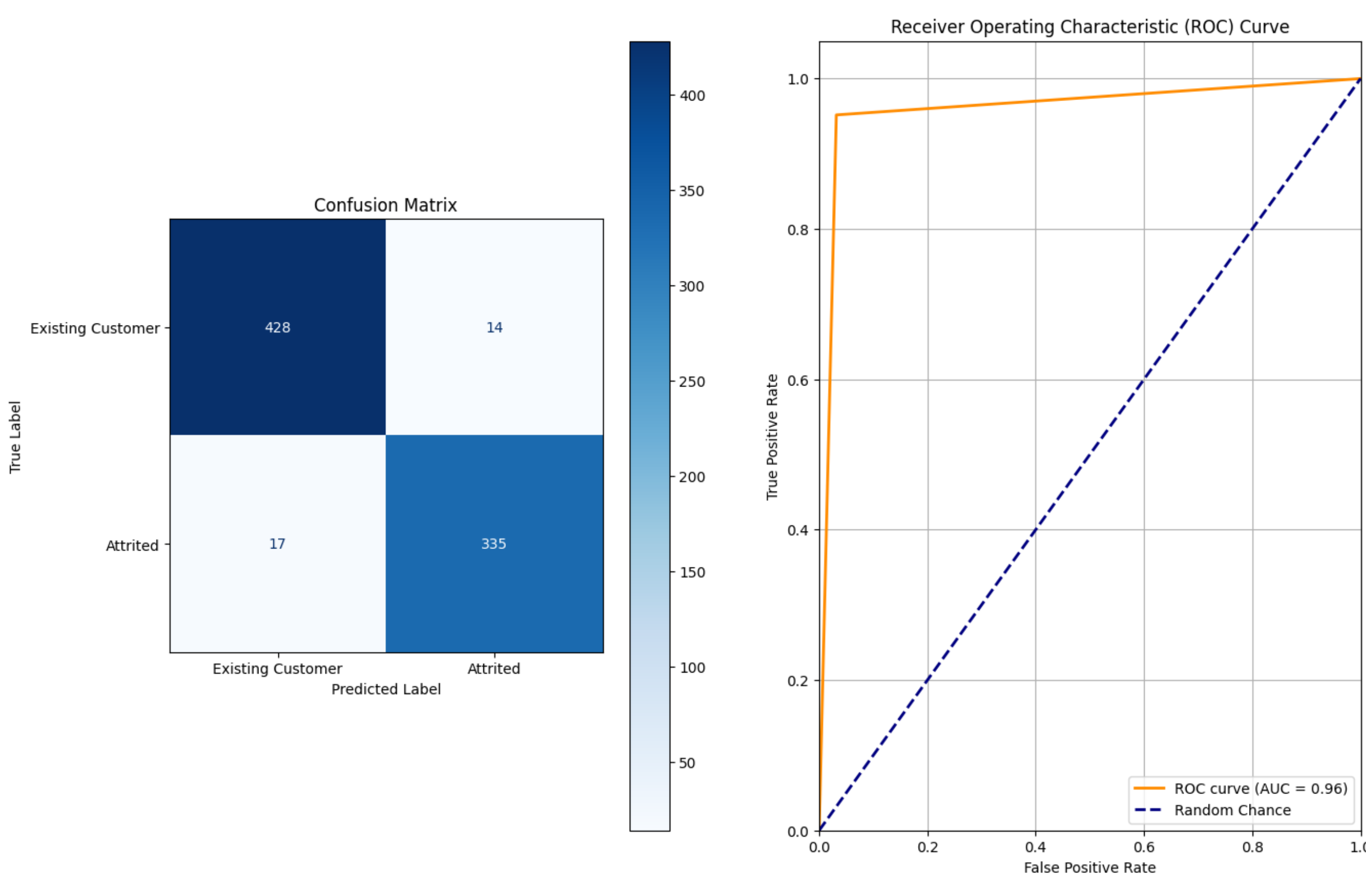


Image 6: Optimized KNN Model Confusion Matrix and ROC Curve

Critical performance metrics for the Optimized Models are found in Table 2, below:

Logistic Regression	92.19%	90.62%	91.67%	0.91
QDA	92.32%	95.17%	88.39%	0.92
KNN	96.10%	95.17%	95.99%	0.96

Table 2: Optimized Model Performance Metrics

An in-depth comparison of model performance can be found in the Model Comparison section.

Robust Variable Selection Process

The variable selection process was performed with a combination of forward and backward stepwise selection. Each subset Ordinary Least Squares model in the combinatorial predictor evaluation was scored using Mallow's C_p, equation seen in Image 7, below.

$$C_p = \frac{1}{n}\left(\mathrm{RSS} + 2d\hat{\sigma}^2\right),$$

Image 7: Equation for Mallow's C_p

where:

n is the number of observations,

σ^2 is an estimate of the variance of the error ε associated with each response measurement,

d is the number of predictors,

RSS is the Residual Sum of Squares of the model's prediction error

Upon completion of the stepwise selection process, the critical variables were determined. Table 3, below, details the selected variables, as well as the initial variables that were selected during the analytic plan phase.

Stepwise Selected Variable	Analytic Plan Initial Variables
Balance_To_TC	Balance_To_TC
Card_Category	Card_Category
Contacts_Count_12_mon	Contacts_Count_12_mon
Credit_Limit_Per_Income	Credit_Limit_per_Income
Dependent_count	**NOT SELECTED**
Education_Level	Education_Level
Gender	Gender
Leveraged_Balance_Ratio	Leveraged_Balance_Ratio
Marital_Status	Marital_Status
Months_Inactive_12_mon	**NOT SELECTED**
Total_Amt_Chng_Q4_Q1	Total_Amt_Chng_Q4_Q1
Total_Ct_Chng_Q4_Q1	Total_Ct_Chng_Q4_Q1
Total_Relationship_Count	**NOT SELECTED**
Total_Revolving_Bal	Total_Revolving_Bal
Total_Trans_Amt	Total_Trans_Amt
Total_Trans_Ct	Total_Trans_Ct
NOT SELECTED	Avg_Utilization_Ratio

NOT SELECTED	Income_Category
NOT SELECTED	Relationship_Strength

Table 3: Stepwise Variable Selection vs Analytic Plan Variable Selection

The similarities between the two selection strategies are likely a result of the analysis completed during the Analytic Plan phase, where variables were scored based on a combination of correlation with the response variable and the outlier count. The Stepwise Variable Selection strategy dropped the following variables: Avg_Open_To_Buy, Avg_Utilization_Ratio, Credit_Limit, Customer_Age, Income_Category, Months_on_book.

As seen in the Model Comparison section, there was a performance increase across every model after feature selection and class imbalance correction. Table 4, below, highlights the improvement in performance metrics for each model from the Null model to the optimized configuration.

Model Type	Accuracy Increase	Recall Increase	Precision	F1-Score
Logistic Regression	2.06%	34.62%	13.58%	0.26
QDA	10.54%	4.31%	41.07%	0.3
KNN	9.89%	59.46%	30.89%	0.5

Table 4: Performance Metric Difference Between Null and Optimized Models

The most notable increases in performance come from the KNN and QDA models. This implies that most of the linearity in the data can be fit by the Logistic Regression model with all variables present. Alternatively, many of the localized and non-linear relationships are revealed through feature selection and class imbalance correction, leading to a dramatic increase in performance for QDA and KNN models.

Model Comparison

After training each model on the respective configurations of features, a summary table of model performance could be generated, as seen in Table 4, below. Note the percent reduction indicated in the Feature Configuration column. This describes the percentage reduction of observations in the training set to mitigate class imbalance.

Feature Configuration	Model Type	Accuracy	Recall	Precision	F1-Score
Null – No Reduction	Logistic Regression	90.13%	56%	78.09%	0.65
Null – No Reduction	QDA	81.78%	90.86%	47.32%	0.62
Null – No Reduction	KNN	86.21%	35.71%	65.1%	0.46
Null – 65% Reduction	Logistic Regression	93.28%	90.40%	89.30%	0.90
Null – 65% Reduction	QDA	91.45%	91.02%	84.24%	0.88
Null – 65% Reduction	KNN	81.47%	60.37%	78.31%	0.68
Null – 75% Reduction	Logistic Regression	91.94%	90.06%	91.62%	0.91
Null – 75% Reduction	QDA	91.56%	90.34%	90.60%	0.90

Null – 75% Reduction	KNN	78.46%	67.05%	81.10%	0.73
Initial Variable Selection – No Reduction	Logistic Regression	87.77%	38.29%	75.71%	0.51
Initial Variable Selection – No Reduction	QDA	87.63%	74.86%	60.09%	0.67
Initial Variable Selection – No Reduction	KNN	89.99%	60.57%	74.13%	0.67
Initial Variable Selection – 65% Reduction	Logistic Regression	92.46%	86.38%	90.29%	0.88
Initial Variable Selection – 65% Reduction	QDA	90.84%	94.12%	81.07%	0.87
Initial Variable Selection – 65% Reduction	KNN	96.64%	95.36%	94.48%	0.95
Initial Variable Selection – 75% Reduction	Logistic Regression	91.56%	88.92%	91.79%	0.90
Initial Variable Selection – 75% Reduction	QDA	91.56%	95.45%	86.82%	0.91
Initial Variable Selection – 75% Reduction	KNN	96.10%	95.17%	95.99%	0.96
Stepwise Selected Variables- No Reduction	Logistic Regression	89.14%	50.00%	76.09%	0.60
Stepwise Selected Variables- No Reduction	QDA	87.58%	76.29%	59.73%	0.67
Stepwise Selected Variables- No Reduction	KNN	90.46%	64.57%	74.34%	0.69
Stepwise Selected Variables- 65% Reduction	Logistic Regression	92.97%	88.54%	89.94%	0.89
Stepwise Selected Variables- 65% Reduction	QDA	90.63%	94.43%	80.47%	0.87
Stepwise Selected	KNN	96.64%	95.05%	94.75%	0.95

Variables- 65% Reduction					
Stepwise Selected Variables- 75% Reduction	Logistic Regression	92.19%	90.62%	91.67%	0.91
Stepwise Selected Variables- 75% Reduction	QDA	92.32%	95.17%	88.39%	0.92
Stepwise Selected Variables- 75% Reduction	KNN	96.10%	95.17%	95.99%	0.96

Table 5: Summary of Model Performance Across Feature and Training Configurations

The highest performing optimized model identified during this evaluation was KNN with a neighbor count of 7. Interestingly, this model produces the same performance metrics for both Initial Variable Selection – 75% Reduction and Stepwise Selected Variables- 75% Reduction feature configurations. The performance metrics for this model are as follows:

Accuracy	AUC	Recall	Precision	F1-Score
96.10%	0.96	95.17%	95.99%	0.96

Table 6: Optimal KNN Performance Metrics

The KNN model likely performs the best on the reduced dataset because of non-linear relationships and localized patterns within the data. Logistic regression will attempt to make a linear approximation of the non-linear relationships, causing a drop in accuracy. QDA models can fit non-linear relationships, but they struggle with decision boundaries that are not smooth curves, as well as localized patterns. QDA outperforms Logistic regression in the optimized feature configurations, which is evidence of non-linear relationships. KNN further outperforming QDA is an indication of both non-linear relationships and localized clusters of patterns within the data.

Across the three models, KNN is the lowest interpretability, however, it produces performance gains that are greater than 4% across all metrics as compared to QDA or Logistic Regression. KNN fits the data well because it learns localized relationships between variables, not one global equation that describes the data like Logistic Regression. Given the business case, accuracy is paramount to solve the problem at hand, therefore KNN is advantageous despite being a “black box” model. In practice, this principle is common, as deep learning models fall further down the interpretability score hierarchy, but they are the flagship models for nearly every domain where artificial intelligence and machine learning are applicable.

Recommendations for Phase 3

Based on the outcome of the model evaluation phase, models with the ability to fit non-linear relationships and localized patterns are advantageous for this data set. To further improve predictive performance on the data, models such as Localized Regression, Gradient Boosted Decision Trees, or Random Forest can be explored. Given the performance of KNN during this phase of analysis, a distance weighted KNN model may provide additional performance benefits.

Additionally, features could be added by arbitrarily combining different quantitative variables with various mathematical operations, then further stepwise selection could be used to identify which combinations increase predictability. This would introduce combinations of variables with high predictive value that may not have been explored otherwise.